모컴 - Mini-project 주제

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Mini Project Proposal (Revised)

1. Goal & Usage Scenario

The goal of this project is to build an **on-device Android app** that estimates whether an expressed emotion is genuine or fake using voice and physiological response.

The app has two modes:

Mode 1 - Voice + Finger PPG (Main Mode)

- We collect our own voice + finger PPG data.
- Voice is recorded through the smartphone microphone.
- Finger PPG is recorded through the smartphone camera and flash.
- This is the main mode because it is easier to collect and does not require visible face recording.

Mode 2 - Voice + Finger PPG + Face (Extended Mode)

- A face model is trained separately using public face datasets.
- At inference time, the face model is attached to the Mode 1 result.
- We test whether face output should be fused or only compared.

Main research question: **Does adding a separately trained face model improve our self-collected voice + PPG model?**

This structure is chosen because collecting genuine/fake facial video data is difficult, privacy-sensitive, and hard to scale. Public face datasets are also limited for this exact task.

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2. Related Works

Voice emotion recognition uses acoustic features such as MFCC, pitch, energy, speaking rate, jitter, and shimmer. These features can reveal stress, hesitation, excitement, or vocal instability.

PPG measures blood volume changes and can estimate heart rate and HRV. We use **smartphone finger PPG** instead of face-based rPPG because rPPG is sensitive to lighting, face movement, camera angle, and skin-region tracking.

Face datasets such as AffectNet, CASME II, and SAMM are useful for facial emotion or micro-expression recognition. However, they are not enough to train a robust genuine/fake model alone. Therefore, face is used as an optional branch, not the main data source.

Project strategy: train the main model with self-collected voice + finger PPG, train the face model with public face datasets, and compare Mode 1 vs. Mode 2.

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3. Key Idea

Overall Architecture

```
Self-collected voice + finger PPG      Public face dataset
              |                                |
              v                                v
      Voice + PPG model                  Face model
              |                                |
              +------------+-------------------+
                           v
                 Compare or late-fuse
```

Mode 1: Voice + Finger PPG

Mode 1 is the primary deliverable.

Input	Features
Voice	MFCC, pitch, energy, jitter/shimmer, speaking rate
Finger PPG	BPM, HRV, peak interval, signal quality

```
Voice features -> voice encoder -> z_voice
PPG features   -> PPG encoder   -> z_ppg
concat(z_voice, z_ppg) -> MLP -> emotion + genuine/fake
```

Mode 1 is practical because voice and finger PPG are easy to collect, privacy-friendly, and more stable than face-based rPPG.

Mode 2: Add Face Model

Mode 2 adds a public-data face model to Mode 1.

```
Public face dataset -> face encoder -> facial emotion probability
```

At inference time:

```
Mode 1 output: voice + PPG -> prediction
Face output:   face        -> facial emotion
```

```
Final step:    compare only OR late fusion
```

Late fusion is preferred because the main model and face model are trained on different datasets. The final score can combine `voice_PPG_score`, `face_score`, and `mismatch_score`.

If the phone cannot use front and rear cameras at the same time, Mode 2 can use sequential capture: record face/voice first, then measure finger PPG before or after the speech clip.

4. Implementation Plan

Dataset

Dataset	Modality	Usage
Self-collected	Voice + smartphone finger PPG	Main Mode 1 training
RAVDESS / CREMA-D	Voice	Optional voice pretraining
AffectNet / FER-style datasets	Face image	Face emotion model
CASME II / SAMM	Micro-expression video	Optional face experiment

Self-Collection Protocol

Each participant records 5-10 second clips while speaking and covering the camera/flash with a finger for PPG. Target scale: 4-6 participants, 20-40 clips per participant.

Example labels:

Scenario	Label
Talk about something you like	genuine positive
Say you like something you dislike	fake positive
Say “I am fine” after a stressful clip	fake neutral
Read or act an emotion from a script	neutral / acted emotion

Feature Extraction

Branch	Processing
Voice	16 kHz audio, silence removal, MFCC, pitch, energy, jitter
Finger PPG	RGB intensity, bandpass filtering, peak detection, BPM/HRV
Face	face crop or MediaPipe Face Mesh landmarks

Finger PPG is preferred over rPPG because it is less sensitive to lighting and face movement.

Model Architecture

Component	Architecture
Voice branch	1D-CNN/LSTM or feature-based MLP
PPG branch	MLP or small 1D-CNN
Face branch	MediaPipe landmarks + MLP/LSTM
Fusion head	MLP classifier

Mode 2 is evaluated in two versions:

Version	Meaning
Mode 2-A	Compare Mode 1 and face outputs separately
Mode 2-B	Late-fuse Mode 1 score and face score

Model Optimization

Because the final system runs on a smartphone, all models are converted to TensorFlow Lite. TensorFlow Lite INT8 post-training quantization is applied to the Mode 1 model, the face model, and the optional fusion classifier. The target is about **75% model size reduction**, **less than 2% accuracy degradation**, and real-time or near real-time Android inference.

If INT8 quantization hurts accuracy too much, float16 quantization or quantization-aware training can be used as a fallback.

Android App

Because finger PPG uses the camera, the app supports simultaneous or sequential capture depending on device support.

Mode 1 - Voice + Finger PPG

```
┌──────────────────────────────┐
│  🎙️ Voice + Finger PPG Mode    │
│                              │
│  😊 Outward emotion: Happy     │
│     [██████░░]  87%          │
│  😠 Hidden emotion: Angry      │
│     [████░░░░]  62%          │
│                              │
│  Deception probability        │
│     [███████░]  78%          │
│  ❤ BPM: 94   HRV: low       │
│  PPG signal quality: stable  │
│                              │
│  ⚠ This emotion may be fake.  │
└──────────────────────────────┘
```

Mode 2 - Voice + PPG + Face

```
┌────────────────────────────┐
│  [Camera feed + landmarks]   │
│                              │
│  😊 Outward emotion: Happy    │
│     [██████░░]  87%          │
│  😠 Hidden emotion: Angry     │
│     [████░░░░]  62%          │
│                              │
│  Deception probability        │
│     [███████░]  78%          │
│  ❤ BPM: 94   HRV: low       │
│  Face-main discordance: high │
│                              │
│  ⚠ Face and main model differ.│
│  ⚠ This emotion may be fake. │
└────────────────────────────┘
```

5. Evaluation Goal

The main evaluation compares Mode 1 and Mode 2.

Metric	Mode 1	Mode 2-A	Mode 2-B
Emotion accuracy	-	-	-
Genuine/fake accuracy	-	-	-
F1-score	-	-	-
Android latency	-	-	-
TFLite model size	-	-	-

Expected conclusion: if Mode 2-B improves accuracy, face adds useful information; if only comparison helps, face should be used as a mismatch signal; if Mode 2 does not help, Mode 1 remains the practical final app.

6. Project Timeline

Period	Task
Until May 4	Finalize proposal and mode definition
May 5 - May 11	Build data collection app and label protocol
May 12 - May 18	Collect voice + PPG data; implement preprocessing
May 19 - May 25	Train Mode 1 and public-data face model
May 26 - June 1	Implement fusion, TFLite conversion, Android UI
June 2 - June 4	Evaluate results and prepare final demo

References

[1] S. R. Livingstone and F. A. Russo, “The Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS),” PLOS ONE, 2018. https://doi.org/10.1371/journal.pone.0196391
[2] H. Cao et al., “CREMA-D: Crowd-Sourced Emotional Multimodal Actors Dataset,” IEEE Transactions on Affective Computing, 2014. https://doi.org/10.1109/TAFFC.2014.2336244
[3] W.-J. Yan et al., “CASME II: An Improved Spontaneous Micro-Expression Database and the Baseline Evaluation,” PLOS ONE, 2014. https://doi.org/10.1371/journal.pone.0086041
[4] A. K. Davison et al., “SAMM: A Spontaneous Micro-Facial Movement Dataset,” IEEE Transactions on Affective Computing, 2018. https://doi.org/10.1109/TAFFC.2016.2573832
[5] A. Mollahosseini et al., “AffectNet: A Database for Facial Expression, Valence, and Arousal Computing in the Wild,” IEEE Transactions on Affective Computing, 2019. https://doi.org/10.1109/TAFFC.2017.2740923
[6] C. Lugaresi et al., “MediaPipe: A Framework for Building Perception Pipelines,” arXiv:1906.08172, 2019.
[7] J. Allen, “Photoplethysmography and its application in clinical physiological measurement,” Physiological Measurement, 2007. https://doi.org/10.1088/0967-3334/28/3/R01
[8] E. Jonathan and M. Leahy, “Investigating a smartphone imaging unit for photoplethysmography,” Physiological Measurement, 2010. https://doi.org/10.1088/0967-3334/31/11/N01
[9] TensorFlow, “TensorFlow Lite model optimization.” https://www.tensorflow.org/lite/performance/model_optimization
[10] TensorFlow, “Post-training quantization.” https://www.tensorflow.org/lite/performance/post_training_quantization